



```
render() {  
  return (  
    <React.Fragment>  
      <div className="py-5">  
        <div className="container">  
          <Title name="our" title="product">  
            <div className="row">  
              <ProductConsumer>  
                {(value) => {  
                  console.log(value)  
                }}  
              </ProductConsumer>  
            </div>  
          </div>  
        </div>  
      </React.Fragment>  
    )  
  )  
}
```

Secure Student Information System (SSIS)

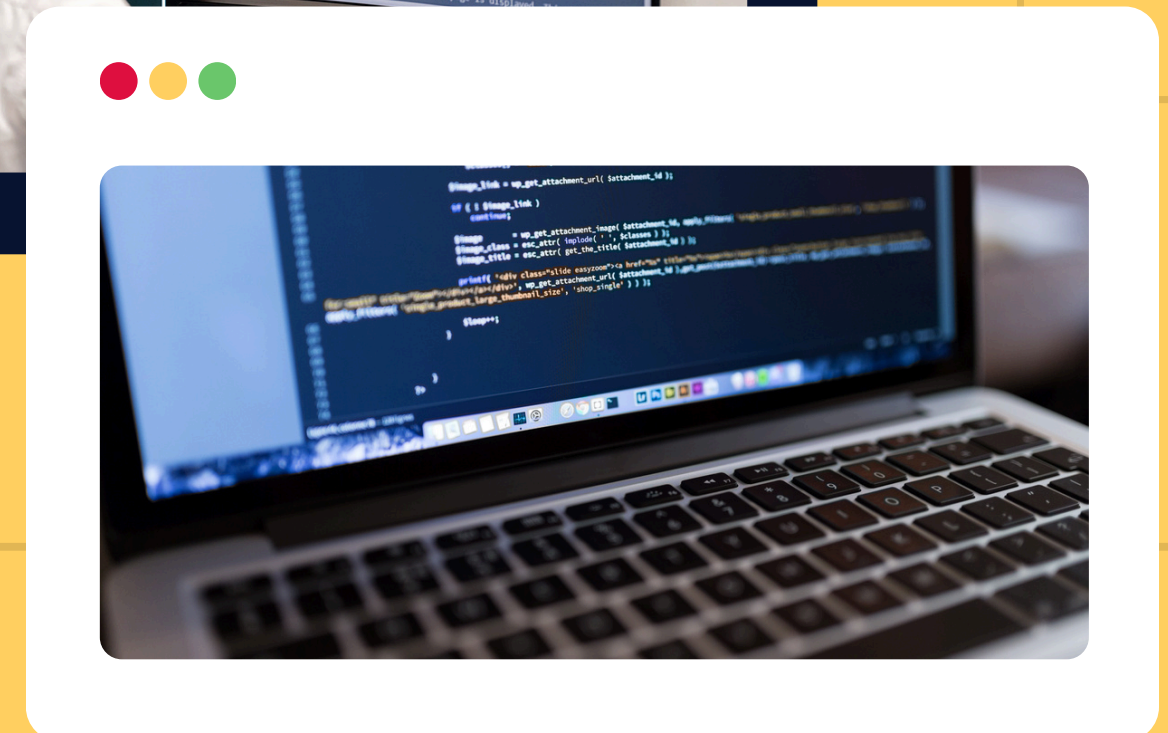
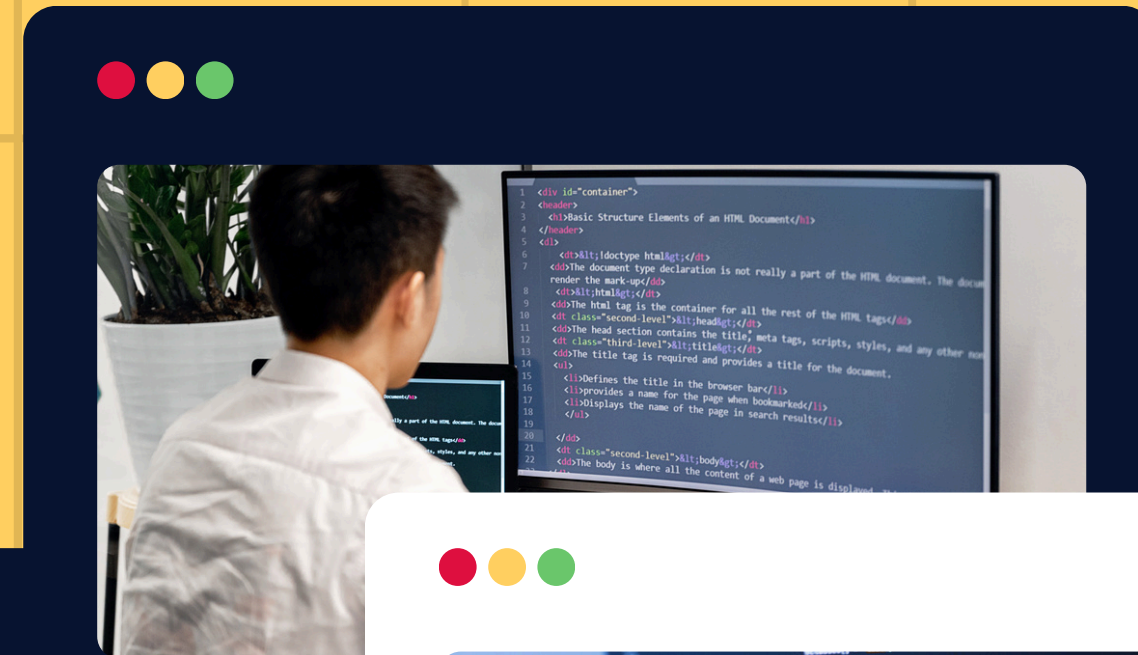
Presented By

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Project Overview

This project is a secure web-based Student Information System developed for the Secure Software Engineering course. The system allows students to securely log in and view their grades, while instructors can post and update grades.

The main goal of the project is to apply secure software engineering principles throughout the Software Development Life Cycle (SDLC) to protect sensitive academic information and prevent common cyberattacks.





Functional Requirements

- Allow students and instructors to create accounts.
- Validate usernames and passwords before authentication.
- Display error messages for invalid login attempts.
- Allow users to update account information.
- Allow users to log in securely.
- Allow instructors to post grades.
- Allow instructors to update grades.
- Allow students to view their grades.

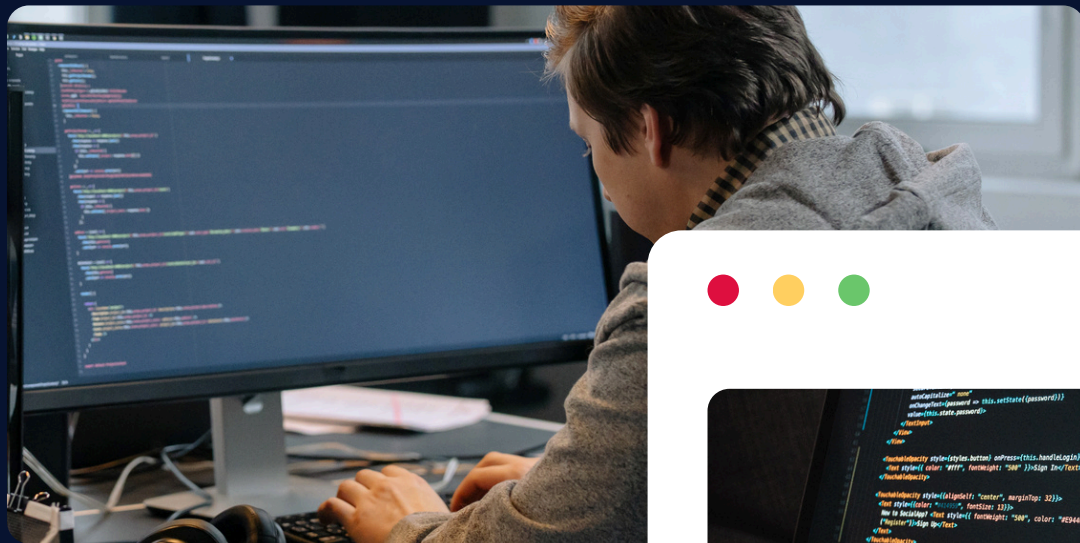


Nonfunctional Requirements

- The system must be secure and protect users' information.
- The system must be available at all times for authorized users.
- The system must have fast response times.
- The system must validate all user inputs.
- The system must prevent unauthorized access.
- The system must maintain the confidentiality and integrity of grades.
- The system must be easy to use for students and instructors.

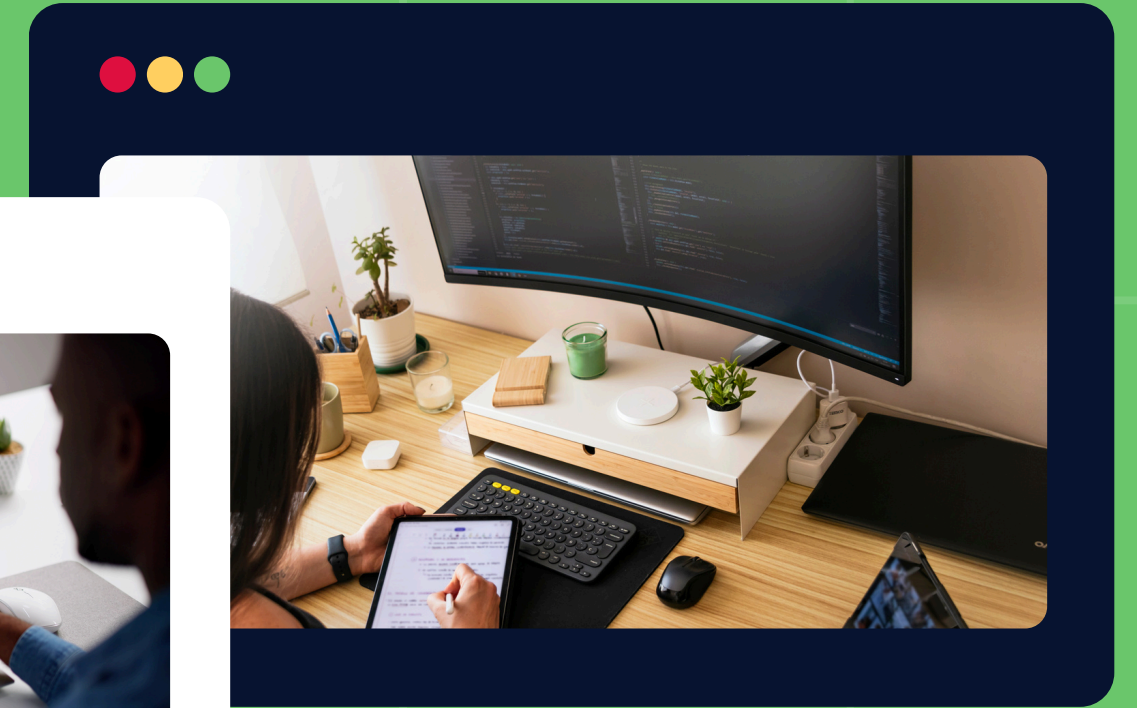
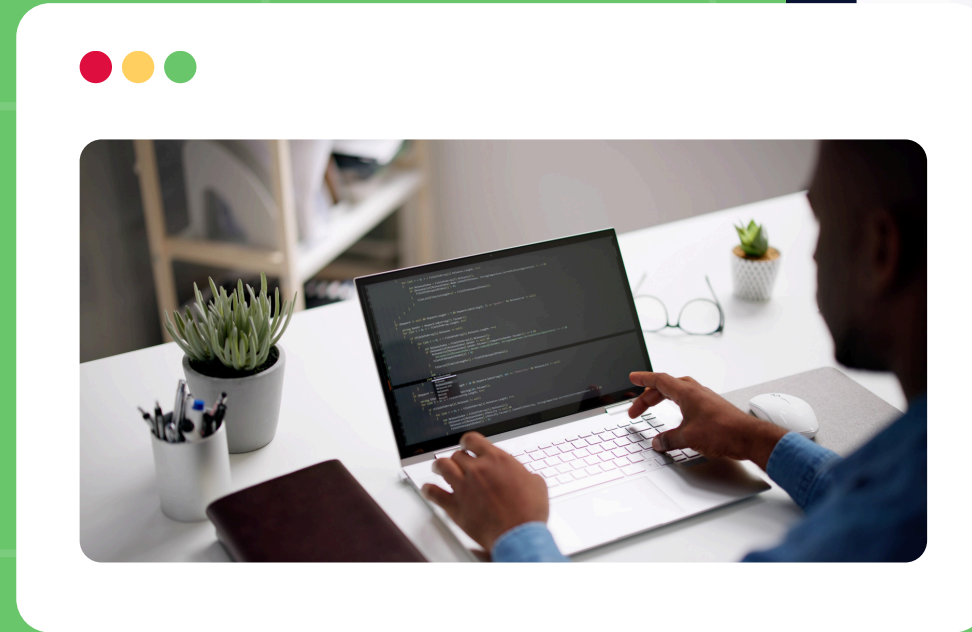
Security Requirements Analysis

Security Requirements



- Student IDs must contain numbers only.
- Student IDs must be exactly 9 digits.
- Passwords must contain between 8 and 20 characters.
- The system must detect possible SQL injection attempts.
- Only authorized users can access the system.
- Students can only view their own grades.
- Instructors can only post and update grades.
- Invalid login attempts must display secure error messages.

Identify Assets & Threats



- Student Data
- Grade Records
- System Credentials
- Availability

- Spoofing
- Tampering
- Information Disclosure
- Denial of Service (DoS)

Risk Matrix

Threat	Likelihood	Impact	Risk Level	Mitigation
SQL Injection	High	High	Critical	Input validation and filtering
Unauthorized Access	Medium	High	High	Authentication and access control
Invalid Input Attack	High	Medium	High	Input validation
Weak Passwords	Medium	Medium	Medium	Password validation

Misuse Cases



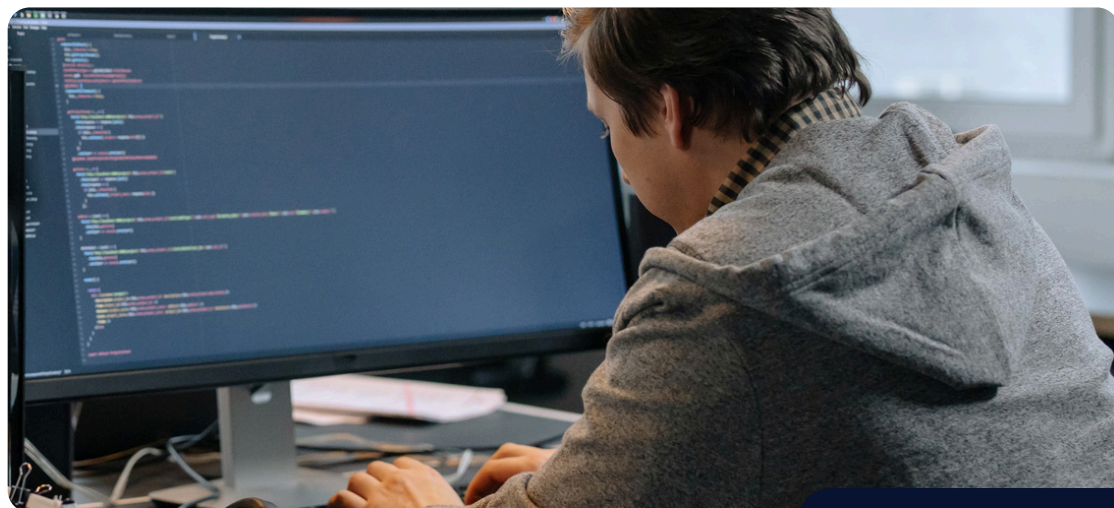
1. A student attempts to inject SQL commands into the login field to bypass authentication.

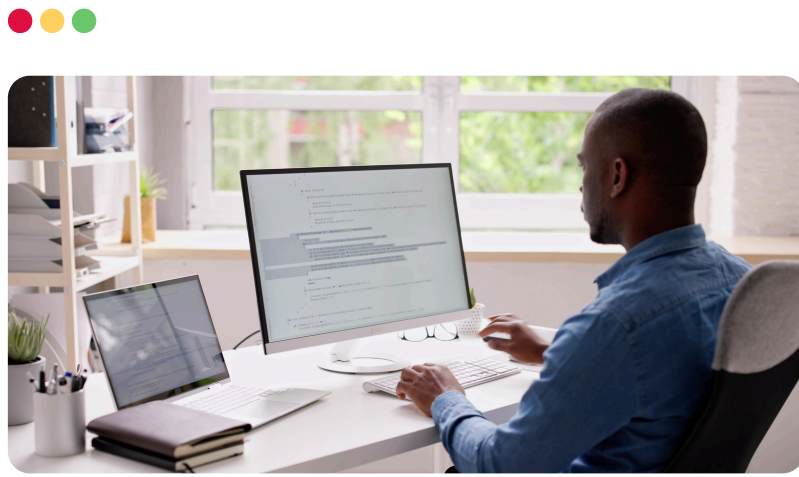


2. A student attempts to access a URL intended for instructors (e.g., /update-grades) by guessing the path.



3. An unauthorized script attempts to automate thousands of login attempts to crack a password.





Securing Design

Architecture & Components



The project follows a three-tier architecture:

1. **Presentation Layer** – User interface and secure communication
2. **Logic Layer** – Authentication, validation, and access control
3. **Data Layer** – Secure database storage and restricted access



Security principles applied include:

- Least Privilege
- Defense in Depth
- Secure Authentication
- Role-Based Access Control (RBAC)

Threat Modeling (STRIDE)



Design Principles Applied



Database Security

- Role-based access control
- Secure storage of student credentials and grades
- Prepared statements prevent SQL injection
- Database backup procedures
- Future enhancement: password hashing and encrypted data



Encryption Techniques

- HTTPS/TLS for secure communication
- Password hashing before storage
- Secure session tokens
- Protects data confidentiality and integrity



Website Security Measures

- Input validation and sanitization
- Session timeout for inactive users
- Secure cookies for session management
- Generic error messages for security



Preventive Measures Against Regular Attacks

- SQL injection prevention through input filtering
- Login attempt restrictions against brute-force attacks
- XSS prevention using input sanitization
- Secure sessions and authentication prevent unauthorized access

Secure Implementation

Secure login using Student ID and password

- Student ID must:
 - Contain numbers only
 - Be exactly 9 digits
- Password must:
 - Be 8–20 characters long

SQL Injection Prevention:

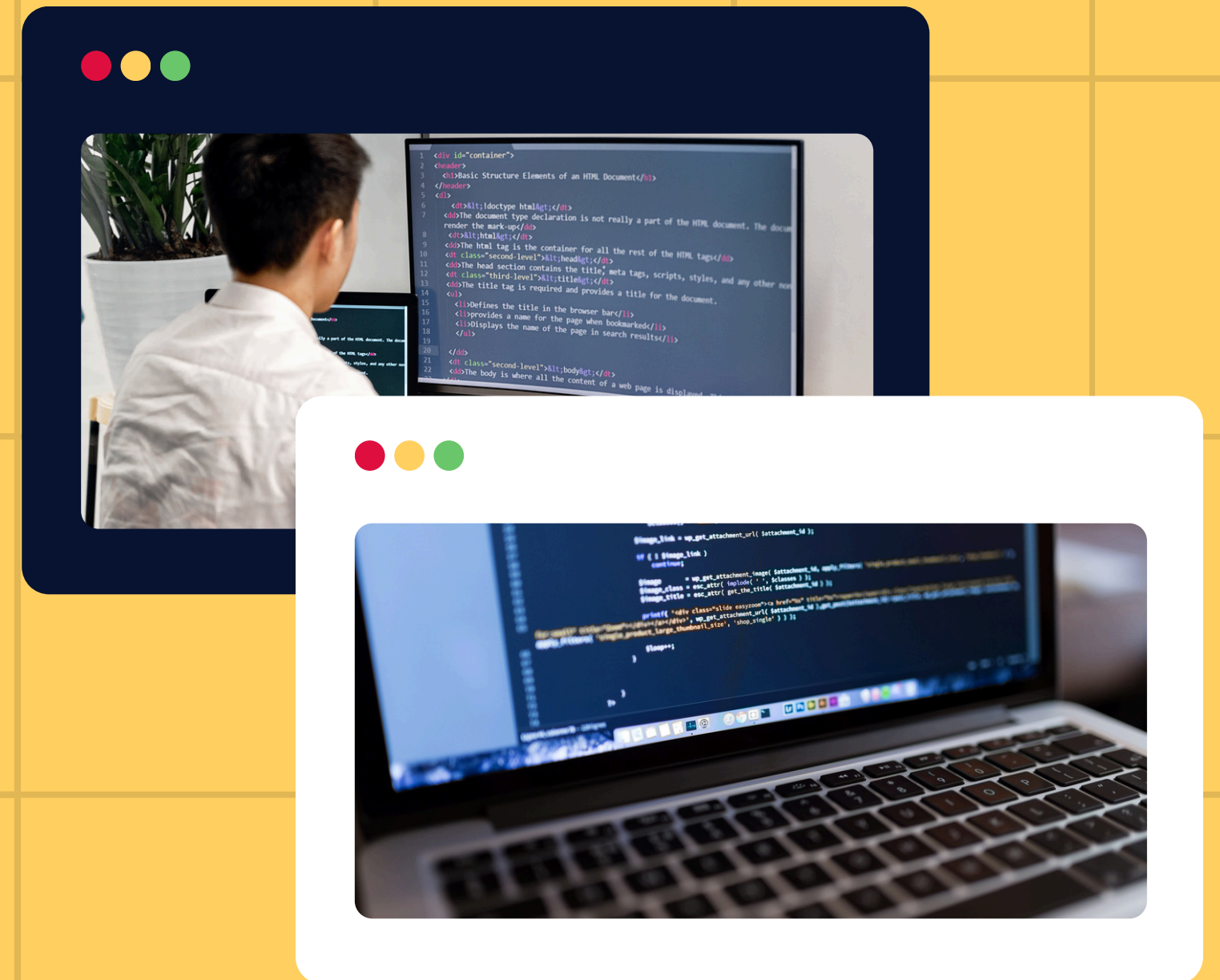
- Detects suspicious inputs (' , =)
- Rejects malicious login attempts

Result: Secure authentication and protection against common attacks.

Defensive Security:

- Validates all user input
- Blocks unauthorized access
- Displays secure error messages

Secure Testing



Test Case 1: SQL Injection Prevention

```
--- exec:3.1.0:exec (default-cli) @ SecureProject ---  
Enter student ID: ' OR '1'='1  
Error: Student ID must only contain numbers  
Enter student ID: 202300112  
Enter password: ' OR '1'='1  
Error: Possible SQL injection detected.  
  
-----  
BUILD SUCCESS  
  
-----  
Total time: 38.838 s  
Finished at: 2026-05-09T14:14:16+03:00  
-----
```

Test Case 2: Length Validation

```
--- exec:3.1.0:exec (default-cli) @ SecureProject ---  
Enter student ID: 2002  
Error: Student ID must be 9 digits!  
Enter student ID: 20120000000000000000  
Error: Student ID must be 9 digits!  
Enter student ID: 202301258  
Enter password: hello  
Error: Password is too short.  
Enter password: helloooooooooooooo123123123  
Error: Password is too long.  
Enter password: hello123123  
Error: Invalid username or password.  
-----  
BUILD SUCCESS  
-----  
  
Total time: 02:48 min  
Finished at: 2026-05-09T14:29:20+03:00
```

Test Case 3: Type Validation

```
--- exec:3.1.0:exec (default-cli) @ SecureProject ---  
Enter student ID: Student22  
Error: Student ID must only contain numbers  
Enter student ID: 20210076!  
Error: Student ID must only contain numbers  
Enter student ID: 202300112  
Enter password: Student@pmu2  
Error: Invalid username or password.  
  
-----  
BUILD SUCCESS  
-----  
  
Total time: 02:24 min  
Finished at: 2026-05-09T14:33:31+03:00
```


Test Case 4: Successful Login

```
--- exec:3.1.0:exec (default-cli) @ SecureProject ---  
Enter student ID: 202201658  
Enter password: cherry50  
Login successful!  
-----  
BUILD SUCCESS  
-----  
Total time: 21.595 s  
Finished at: 2026-05-09T14:35:41+03:00
```

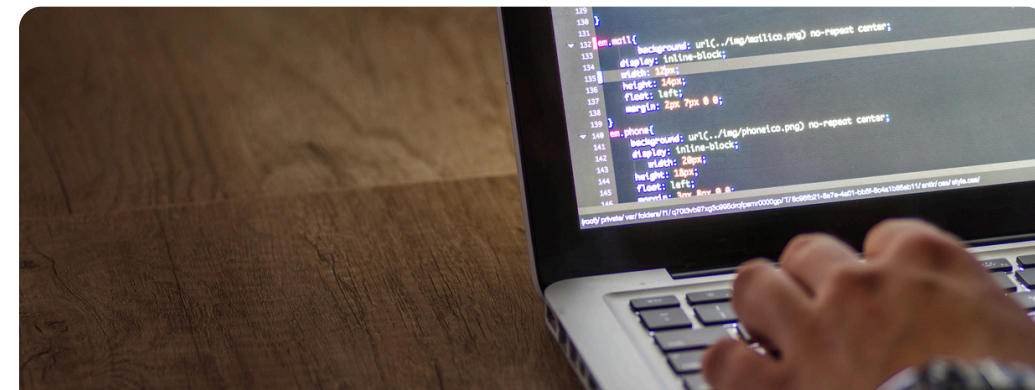
```
--- exec:3.1.0:exec (default-cli) @ SecureProject ---  
Enter student ID: 202301258  
Enter password: clouds@22  
Login successful!  
-----  
BUILD SUCCESS  
-----  
Total time: 12.445 s  
Finished at: 2026-05-09T14:37:11+03:00
```

Conclusion



This project demonstrates how security can be integrated throughout the SDLC to build a secure and reliable student information system. By applying secure coding practices, validation techniques, and access control mechanisms, the system protects sensitive academic information and reduces common security vulnerabilities.





Thank You